Where a fact lives: synaptic memory in Dragon Hatchling

Concept: Synaptic plasticity as short-term memory · **Artifact:** an interactive explainer with a live 27,776-parameter BDH-GPU running in the browser

The design pressure. A Transformer remembers what it has read by keeping it: the KV cache grows linearly with context, so memory cost tracks how much has been said rather than how much matters. Eviction, compression, retrieval, linear attention and state-space models all try to hold within-session memory in something of fixed size. That buys bounded cost and forces a question the Transformer never has to answer: when two facts compete for the same finite state, what happens to them?

What BDH changes. Dragon Hatchling (BDH) reformulates attention as a *synaptic write*. Its GPU formulation carries a state ρ updated by

```
ρ_{t,l} := (ρ_{t-1,l} + LN(E y_{t,l-1}) x_{t,l}■) U — Eq. (8), arXiv:2509.26507
```

Each token deposits a rank-1 outer product: reading strengthens a specific set of connections, and the model answers by reading that state back. Two details are routinely garbled. First, ρ is $d \times n$ — in our model 32 × 256 — **not** neuron-by-neuron; the conceptual σ of Eq. (6) relates to it by $\sigma \approx D_y \rho$, rank at most d. Second, U is defined as "local rotation *or damping*"; the public reference implements rotation only (RoPE), so BDH as released has no forgetting term. Activations are sparse and non-negative — the paper reports about 5% of the y vectors active. The trade is that nothing is addressable by position: there is no cache entry to evict, only superposition in one fixed tensor.

	Softmax + KV cache	Linear attention	BDH synaptic state
Memory growth	linear in tokens	fixed-size state	fixed-size state
Where session memory lives	cached keys/values	accumulated state matrix	ρ, synaptic writes over (channel, neuron)
Failure mode	eviction, context limit	interference	interference; positional nulls (measured here)
Interpretability handle	attention maps over tokens	state matrix	individual synapses; reported monosemantic at concept level

Our claim, and the control that carries it. A fact the model just read is held in a small, *locatable* set of synapses rather than in its weights. We train a 27,776-parameter BDH-GPU on multi-query associative recall (Arora et al., *Zoology*, arXiv:2312.04927), resampling name–city pairings uniformly per sequence — so the weight-optimal prior is uniform and the answer cannot come from the parameters. Deleting the binding from context drops recall to 7.5% against a 6.25% chance line; rebinding it moves the answer 99.75% of the time.

Ablating the **8 largest-magnitude entries of the Hebbian write** that laid one binding down — 0.098% of an 8,192-entry state — takes recall from 99.8% to **77.2%** [72.8, 81.0]. The result means nothing without a control, so we remove the largest-magnitude entries *outside* that write: this necessarily removes at least as much state mass, here **2.06×** as much, and recall stays at **100.0%** [99.0, 100.0]. Across a dose ladder, targeted ablation falls to 10.8% while that control never drops below 78%. Untouched bindings in the same sequence fall only 92.4% → 88.0%. The effect is therefore about *which* state was removed, not how much. This is the mechanism the BDH paper reports at §6.3, where in-context state "localize[s] on the same synapses consistently across multiple prompts".

BDH-CQ's role, labelled. BDH-CQ (arXiv:2608.09888) carries the same idea one level up: a contextual memory S_t that demonstrations update at inference, and a latent workspace H_r iterated to answer — no parameter updates, no verbalised chain of thought. Its controlled interventions make our point at task scale: with byte-identical held-out inputs, adding one demonstration at the test complexity lifts depth-five nesting from 19/24 to 24/24, and recovers 13/24 from a 0/24 baseline at ordering length eight. What is in state decides. Label the evidence precisely: its 150M configuration reaches 29.5% pass@2 on public ARC-AGI-1 at a computed $0.00070/task — a **cost-efficiency** result, not an accuracy win, since GPT-5.6 Luna (Low) scores 34.2% on the same set; the audit reproducing 29.5% was run by **co-authors**, so it is not an external reproduction; and capability is structured rather than scalar — rotation composes with relocation on 72/72, colour swap on 0/72. BDH-CQ's dimensions and update rules are proprietary, and nothing we built reproduces them.

The most important limitation. We set out to show both halves of a claim — that the binding is localised *and* that competing facts erase it — and only the first survived. Our interference curve failed its own control: holding the number of competing bindings fixed and varying only neutral filler still moved recall from 50.4% to 100%, so the load axis was confounded with sequence geometry. It is withdrawn. Two further results did not stand: damping never produced the stability–plasticity crossover it predicts, and a capacity ordering across state widths was confounded by training quality. Our model also has an exact failure mode — retrieval fails in periodic bands of query-to-binding offset, at 3 and 13–17 — whose mechanism (RoPE phase cancellation) remains a hypothesis. Beyond this project, the open problem is consolidation: BDH's fast state is erased between sessions, and turning useful fast state into durable slow weights is unsolved. Our model is our own small training run of the published architecture on a synthetic task — not an official Pathway model, and far below any reported BDH scale. It is evidence about the mechanism, not a reproduction of Pathway's results.

Continue with arXiv:2509.26507 §3.2 and §6.3, arXiv:2608.09888 §6, the reference implementation at `github.com/pathwaycom/bdh`, and Arora et al. (arXiv:2312.04927) for the recall task.

Artifact and source: `github.com/Ritik650/DataForge-Pathway`*. Measured values carry Wilson 95% intervals and an explicit n; withdrawn results are documented, not deleted.*